

A Report
On
Architecture Definition / System level Design
of Delta-Sigma Analog to Digital Converter for
Thermal Sensor Applications.

By
Shivakshit Patri
2012AAPS168H

At

INTEL, BANGALORE



A Practice School-II station of



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

December, 2015

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Acknowledgements

The internship opportunity I have with Intel Corporation is a great chance for learning and professional development. Therefore, I consider myself as a very lucky individual as I was provided with an opportunity to be a part of it. I am also grateful for having a chance to meet so many wonderful people and professionals who led me through this period.

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BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE
PILANI (RAJASTHAN)

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Title of the Project: "Architecture definition / System level design of a sigma delta ADC for thermal sensor applications"

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Key Words: Delta Sigma ADC, CIC filter, Thermal Sensor, Thermal Diode

Project Areas: Design of Behavioral Model of Sigma Delta ADC.

Abstract:

The Project focuses on design of a behavioral Model of Sigma Delta ADC for Thermal Sensor Applications using Cadence Virtuoso and Analog-AMS/A. It also Covers the design of behavioral model of Pre-ADC and Post-ADC i.e. the sense Stage and Digital Filter/Decimator. This model is used to develop more detailed Models of Thermal Sensor using Standard Cells which basically sums up the Schematic Design of a Thermal Sensor IP. All the components from Differential Amplifier to Thermal Diode, from Digital Filter to Analog Multiplexers, Are behaviorally modelled using Verilog-AMS

Shivakshit Patri,
Date: 1st December, 2015

Ms. Swapna Kulkarni
Date:

**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE
PILANI (RAJASTHAN)
PRACTICE SCHOOL DIVISION**

Response Option Sheet

Station: INTEL INDIA
ID No: 2012AAPS168H

Center: Bangalore
Name: Shivakshit Patri

Title of the Project: Architecture definition / System level design of a sigma delta ADC for thermal sensor applications

Code No.	Response Option	Course No.(s) & Name
1.	A new course can be designed out of this project.	YES
2.	The project can help modification of the course content of some of the existing Courses	YES
3.	The project can be used directly in some of the existing Compulsory Discipline Courses (CDC)/ Discipline Courses Other than Compulsory (DCOC)/ Emerging Area (EA), etc. Courses	YES (CDC: ADVD)
4.	The project can be used in preparatory courses like Analysis and Application Oriented Courses (AAOC)/ Engineering Science (ES)/ Technical Art (TA) and Core Courses.	NO
5.	This project cannot come under any of the above mentioned options as it relates to the professional work of the host organization.	NO

Shivakshit Patri
Date: 1st December, 2015

Ms. Swapna Kulkarni
Date:

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1. INTRODUCTION

1.1 Thermal sensor :

A device that detects temperature. Thermal sensors are found in many laptops and desktop PCs in order to sound an alarm when a certain temperature has been exceeded. It basically converts the thermal energy of the ambience into much easily measurable electrical energy. There are several kinds of thermal sensors:

1. RTDs (Resistance Temperature Detectors).
2. Thermocouples.
3. Thermistors.
4. Infrared Sensors.

But the project aim is to design a thermal sensor for sensing temperature in laptops and desktops. Therefore all the above thermal mentioned sensors are not suitable here as area/size is a constraint here.

Therefore we use a Thermal Diode for sensing the temperature in our application.

Wikipedia page on Thermal Diode:

This application of thermal diode is based on its property to change voltage across it linearly according to temperature. As the temperature increases, diodes forward voltage decreases. Microprocessors having high clock rate encounter high thermal loads. To monitor the temperature limits thermal diodes are used. They are usually placed in that part of the processor core where highest temperature is encountered. Voltage developed across it varies with the temperature of the diode. All modern Intel CPUs have on-chip thermal diodes. As they are right there in the middle it provides most relevant CPU temperature readings. The silicon diodes have temperature dependency of $-2mV$ per degree Celsius. Thus we can determine the junction temperature by passing a current through the diode and then measuring voltage developed across it. In addition to processors, the same technology is widely used in dedicated temperature sensor IC's.

1.2 Basic Principles

As almost every device characteristic in an integrated circuit is temperature dependent, there are numerous ways of making integrated thermal sensors. Thermal diodes can be implemented by

1. MOS transistors.
2. Bi-polar Transistors.
3. Thermal Delay Lines
4. Tunneling Diodes.

The output of a thermal diode is an analog signal: A temperature-dependent voltage/current's magnitude/frequency.

From Precision Temperature Sensors in CMOS Technology by Michiel A.P. Pertijs:

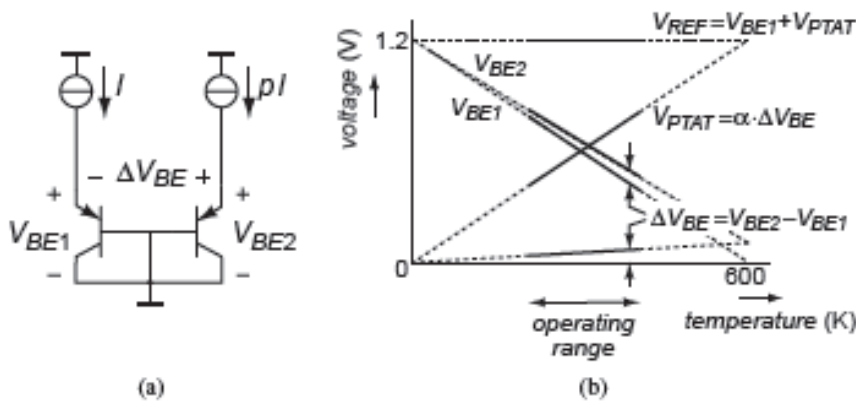


Figure 1.2. Operating principle of a CMOS smart temperature sensor: (a) two diode-connected pnp transistors are biased at a well-defined current density ratio 1 : p; (b) their base-emitter voltages are used to generate a voltage proportional to absolute temperature V_{PTAT} and a bandgap reference V_{REF} , the ratio of which is a measure of temperature.

Most smart temperature sensors make use of the characteristics of bipolar transistors. These characteristics are based on two voltages that can play the role of the signals required for ratio metric temperature measurement: the thermal voltage kT/q (where k is Boltzmann's constant, T is the absolute temperature, and q is the charge of an electron) and the silicon bandgap voltage V_{g0} . The thermal voltage can be used to generate a voltage V_{PTAT} that is proportional to absolute temperature (PTAT), while the bandgap voltage is the basis for generating a temperature-independent reference voltage V_{REF} .

In an integrated smart temperature sensor, a number of bipolar transistors are combined with precision interface circuitry in an analog front-end to extract these voltages. A digital representation of their ratio is then determined by an analog-to-digital converter (ADC). This ratio is a measure of the chip's temperature and is communicated to the outside world (e.g. to a microcomputer) by means of a digital interface.

Figure 1.2 illustrates the operating principle of the analog front-end. In CMOS technology, the bipolar transistors are usually substrate pnp transistors, which are parasitic devices that are present as a side-effect of the MOS transistors for which the technology was designed. The PTAT voltage is generated from the difference in base-emitter voltage ΔV_{BE} between two such transistors biased at different current densities (Figure 1.2a). If the ratio p of the bias currents is well-defined, this difference is accurately PTAT. It is, however, quite small ($0.1-0.25\text{mV/K}$) and therefore needs to be amplified to get a useful voltage V_{PTAT} . The reference voltage is based on the absolute base-emitter voltage of a bipolar transistor, rather than on a difference.

Extrapolated to 0K , this base emitter voltage is equal to the silicon bandgap voltage of about 1.2V (Figure 1.2b). From there, it decreases by about 2mV/K . To compensate for this decrease, a voltage $a \cdot \Delta V_{BE}$ is added to it, resulting in a voltage V_{REF} that is essentially temperature-independent. Since V_{REF} is nominally equal to the silicon bandgap voltage, such a reference is referred to as a 'bandgap reference'.

1.3 Context of the Research (Literature Review)

General-purpose bandgap references were invented before temperature sensors based on the same principles came around. The first bandgap reference was introduced by Hilbiber in 1964 and made practical in the seminal work of Widlar in 1971. The first analog sensor that used ΔV_{BE} as a measure of temperature was presented by Verster in 1968, while the first integrated version of such a sensor was described by Dobkin. A lot of pioneering work was done by Meijer. The first *smart* temperature sensors (based on a ratio metric measurement and with a digital interface) were introduced around 1985 and targeted the relatively small temperature range required to measure human body temperature. They were fabricated in bipolar technology, which dominated the industry at the time. A more general-purpose smart sensor was introduced by Meijer in 1989. During

the 1990's, CMOS technology became mainstream. Due to the ever increasing demand for digital circuits, CMOS technology became available at much lower costs than bipolar technology. Though digital CMOS technology is typically not designed to include bipolar transistors, such transistors usually are available as parasitic devices. These devices come in two flavors, lateral and vertical, both of which have, for general purposes, a poorer performance than their counterparts in bipolar technology. Already in 1983, Vittoz investigated the characteristics and application of lateral bipolar transistors and pointed out that they could be used in CMOS temperature sensors. The first CMOS smart temperature sensor based on lateral bipolar transistors was published by Krummenacher in 1990. The temperature dependency of vertical bipolar transistors has been studied by Wang, who has shown that they are very suitable for making bandgap references and temperature sensors. Moreover, Fruett and Creemer have shown in their study of the piezo-junction effect that vertical pnp transistors in CMOS are relatively insensitive to mechanical stress. This makes them the device of choice, especially for temperature sensors that are exposed to stress due to low-cost plastic packaging. One of the first CMOS smart temperature sensors based on vertical bipolar transistors was presented by Bakker in 1996. His work focused on solving the problem that amplifiers in CMOS technology have a much larger offset than their counterparts in bipolar technology, which prevents accurate amplification of ΔV_{BE} . With his nested-chopper technique, he reduced the offset to the 100 nV level, which reduces the associated temperature errors to negligible levels. The dynamic element matching technique, introduced by Klaassen and Van der Plassche, can be employed to make amplifiers with an accurate gain. Its applicability to temperature sensors has been demonstrated by Meijer *et al.*

Most recent work on smart temperature sensors has been performed in industry, fueled by their above-mentioned application in PCs and laptops. The biggest players in this field are Analog Devices, National Semiconductor and Maxim, who have all developed an extensive line of thermal management chips. For this application, an accuracy of ± 1.0 °C is usually sufficient, although a higher accuracy for the same price is of course a competitive advantage. Developments have mainly focused, however, on incorporating extra functionality on the sensor chip, such as voltage monitoring and temperature measurement of a remote diode located on the microprocessor chip.

2. Methodology

2.1 Behavioral Modelling or System level Design

From Wikipedia:

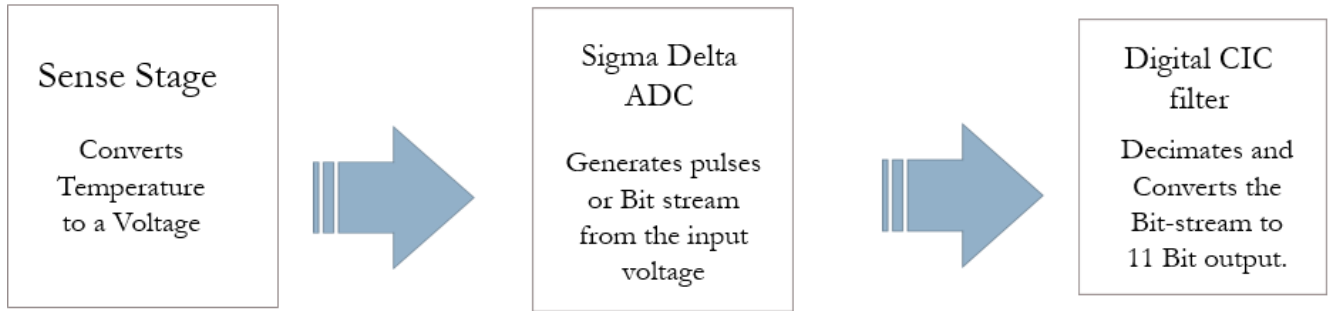
Electronic system level (ESL) design and verification is an emerging electronic design methodology that focuses primarily on the higher abstraction level concerns. The term *Electronic System Level* or *ESL Design* was first defined by Gartner Dataquest, an EDA-industry-analysis firm, on February 1, 2001. It is defined in the *ESL Design and Verification* book as: "*the utilization of appropriate abstractions in order to increase comprehension about a system, and to enhance the probability of a successful implementation of functionality in a cost-effective manner.*"

Electronic System Level is now an established approach at most of the world's leading System-on-a-chip (SoC) design companies, and is being used increasingly in system design. From its genesis as an algorithm modeling methodology with 'no links to implementation', ESL is evolving into a set of complementary methodologies that enable embedded system design, verification, and debugging through to the hardware and software implementation of custom SoC, system-on-FPGA, system-on board, and entire multi-board systems.

The behavioral approach to systems theory and control theory was initiated in the late 70's by J. C. Willems as a result of resolving inconsistencies present in classical approaches based on state-space, transfer function, and convolution representations. This approach is also motivated by the aim of obtaining a general framework for system analysis and control that respects the underlying physics. The main object in the behavioral setting is the behavior --- the set of all signals compatible with the system. An important feature of the behavioral approach is that it does not distinguish a priority between input and output variables.

The Aim of this project is to design a system level circuit/Schematic of the Thermal sensor.

2.2 Thermal Sensor Architecture



The Thermal Sensor in this Project consists of three primary stages:

- The Sense Stage.
- The Analog to Digital Converter.
- The Digital Filter/Decimator.

The Sense Stage consists of the thermal diodes and additional circuitry that switches the input voltage between ΔV_{be} and $-V_{be}$. The Sense stage also consists of the circuitry that rises the input voltage by a constant dc voltage to ensure that the next stage (ADC) works around the common-mode voltage (To ensure that the gain of the ADC's Integrator is linear).

The Analog to Digital Converter used in our Thermal Sensor works on the principle of Delta-Sigma modulation. The double ended differential output from the sense stage is fed as an input to the ADC. The ADC designed here is a 2-stage delta sigma ADC. It contains two integrators (Differential Amplifiers) and one comparator. The output of the comparator is a single bit stream of ones and zeros which is further fed to the digital filter. The other inputs to the ADC include clock (multiple clocks of same frequency but opposite phase).

The functionality of the digital filter is to:

- Decimate the over-sampled bit-stream from the output of the ADC.
- Convert the output of the ADC to an 11-bit binary output.

The detailed architecture of each of the stages will be discussed in the further sections.

.2.1 The Sense Stage

.2.1.1 Temperature characteristics of thermal diode (bipolar junction diode)

The temperature dependency of the base-emitter voltage V_{BE} of a bipolar transistor in its forward-active region depends on the temperature dependency of the saturation current I_S . Moreover, if the transistor is diode-connected and therefore biased via its emitter, the temperature dependency of the common base current-gain a_F is also important. This section describes the temperature dependencies of I_S and a_F , and their effect on V_{BE} .

The final relation between V_{BE} and temperature is given by the equation:

$$V_{BE}(T) = V_{g0} \left(1 - \frac{T}{T_r} \right) + \frac{T}{T_r} V_{BE}(T_r) - \eta \frac{kT}{q} \ln \left(\frac{T}{T_r} \right) + \frac{kT}{q} \ln \left(\frac{I_C(T)}{I_C(T_r)} \right)$$

Where $V_{BE}(T_R)$ is the base-emitter voltage at the specified reference temperature ($T_R = 300\text{K}$). It can be obtained from the spice model parameter of the thermal diode in the name: - tnom.

η is the process variation factor. It can be obtained from the spice model parameter of the thermal diode in the name: - xti.

k is the Boltzmann's constant.

About $1.3806485 \times 10^{-23} \text{ m}^2 \text{ kg s}^{-2} \text{ K}^{-1}$.

q is the charge of a single electron.

About $1.60217662 \times 10^{-19} \text{ coulombs}$.

The product of k and q is around $0.0009862 \text{ V T}^{-1}$.

V_{g0} is the silicon bandgap voltage. It can be obtained from the spice model parameter of the thermal diode in the name: - eg.

$I_C(T_R)$ is the collector current at the specified reference temperature. I_C at any temperature can be represented as $\alpha_F \times I_E$. Where I_E is the emitter current. By maintaining constant I_E at all temperatures (bias current) we can reduce the non-linear dependency of V_{BE} on Temperature. Moreover α_F can be represented as

$$\alpha_F(T) = \frac{\beta_F(T)}{\beta_F(T) + 1}$$

Where β_F is the common emitter current gain of the Bjt. The temperature dependence of β_F is given by

$$\beta_F(T) = \beta_{F0} \left(\frac{T}{T_r} \right)^{X_{TB}}$$

The values of β_{F0} and X_{TB} can be found in the spice model of the thermal diode under the name bf and xtb. (Note: This equation is not considered in our model. Instead β_F is assumed to be close to 1.3 at all temperatures. This has very subtle effect on the accuracy as V_{BE} is a logarithm function of β_F).

V_{PTAT} is generated from the difference in base-emitter voltage ΔV_{BE} between two such transistors biased at different current densities. If the ratio p of the bias currents is well-defined, this difference is accurately PTAT. The relation between ΔV_{BE} and the temperature is given by

$$\Delta V_{BE}(T) = \frac{kT}{q} \ln(p).$$

Where p is the ratio of the bias current densities of the bjts.

At reference temperature ΔV_{BE} is around 40mV and V_{BE} is around 600mV. V_{BE} decreases approximately linearly with temperature, with a slope of about $-2\text{mV}/^\circ\text{C}$.

The below given graph clearly portrays the relation between ΔV_{BE} , V_{BE} and Temperature.

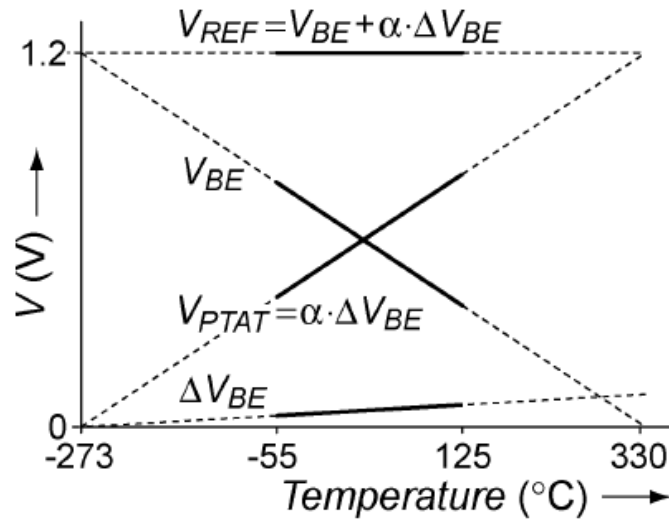
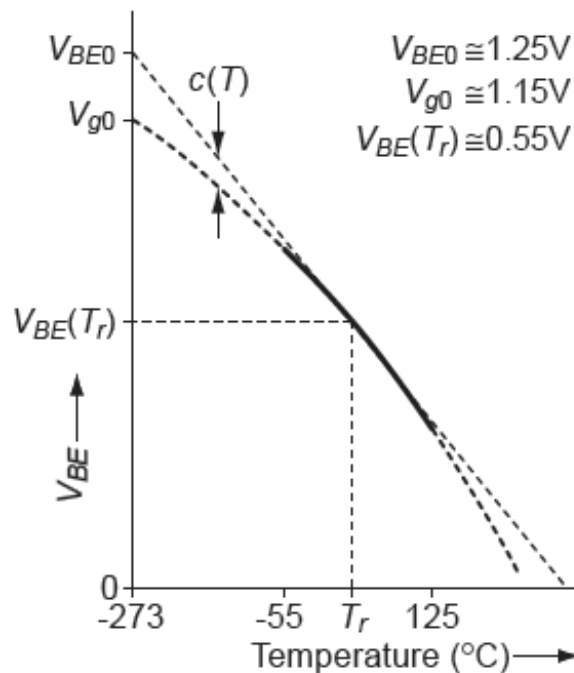


Fig. 2. Temperature dependency of the key voltages in the sensor.

Although ΔV_{BE} is pretty linear with respect to the temperature, V_{BE} is not. A more accurate temperature dependency of V_{BE} is shown below.



The non-linearity increases at both the extremes of the temperature. This non-linearity is due to the process spread factor η .

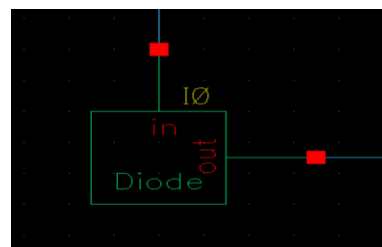
2.2.1.2 Verilog-A code for the Thermal Diode.

// VerilogA for thermal_sensor3, sense_stage, veriloga

```
`include "constants.vams"
`include "disciplines.vams"

module sense_stage(in,out);
input in;
output out;
electrical in,out;
parameter real Temp = 300 from [0:inf);
parameter real VG0 = 1.1 from [0:inf);
parameter real Temp_ref = 298.15 from [0:inf);
parameter real Vbe_ref = 760m from [0:inf);
parameter real Ic_ref = 50u from [0:inf);
parameter real eta = 2.72 from [0:inf);
parameter real beta = 1.3 from [0:inf);
real a = 0.0000862;
//a = K (Boltzmann's constant)/q (charge of an electron)
analog begin
    V(out) <+ VG0*(1-(Temp/Temp_ref)) +
(Temp/Temp_ref)*Vbe_ref - eta*a*Temp*ln(Temp/Temp_ref) +
a*Temp*ln((beta/(beta+1))*I(in)/Ic_ref);
end
endmodule
```

The symbol of this behavioral model:



2.2.1.3 Switching between ΔV_{BE} and $-V_{BE}$

The Delta-Sigma ADC takes basically consists of integrator and a comparator. The integrator comprises of a integrating capacitor and this capacitor keeps charging and discharging based on the input voltage. The input voltage should be switched between ΔV_{BE} and $-V_{BE}$ according to the output of the ADC (bit stream). The reason for this switching will be explained in the later sections of this report. But let's focus on switching circuitry that enables this functionality. A simple analog Multiplexer would suffice for this functionality. Since our project is a behavioral model we need not focus on how to realize an analog MUX. Here is the Verilog-A code for an analog MUX.

```
`include "discipline.h"
`include "constants.h"

//-----
// analog_mux
//
// - analog multiplexer
//
// vin1,vin2:  [V,A]
// vsel:  selection voltage [V,A]
// vout:  [V,A]
//
// INSTANCE parameters
//  vth = 1->0 threshold voltage for the selection line [V]
//
// MODEL parameters
//  {none}
//
// When 'vsel' > 'vth', the output voltage follows 'vin1'
// When 'vsel' < 'vth', the output voltage follows 'vin2'
```

```

module analog_mux(vin1, vin2, vsel, vout);
input vin1, vin2, vsel;
output vout;
electrical vin1, vin2, vsel, vout ;
parameter real vth = 2.5;

integer selector;
real vout_val;
analog begin
    selector = (V(vsel) > vth) ? 1: 0;

    @ (cross(V(vsel) - vth, 1, 1.0, vsel.potential.abstol))
        selector = 1;

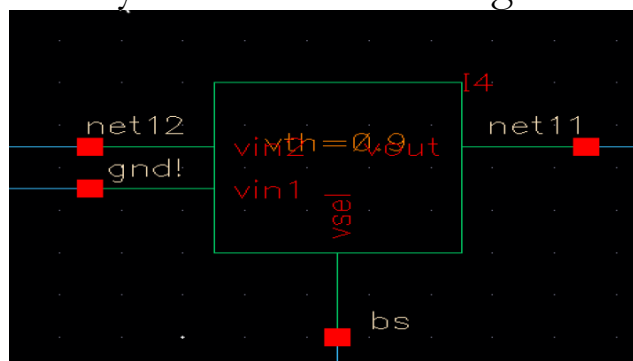
    @ (cross(V(vsel) - vth, -1, 1.0, vsel.potential.abstol))
        selector = 0;

    if (selector == 1)begin
        vout_val = V(vin1);
    end
    else if (selector == 0)begin
        vout_val = V(vin2);
    end

    V(vout) <+ vout_val;
end
endmodule

```

The symbol for this analog MUX:



2.2.1.4 The Biasing Circuit (Common-Mode Voltage)

Since the input to the ADC is an input to the 1st integrator, the input needs to be biased at the required common – mode voltage (0.9V). Again a behavioral Verilog-a model of this circuitry is made and used in the schematic. The Verilog-A code for this biasing circuit is given below:

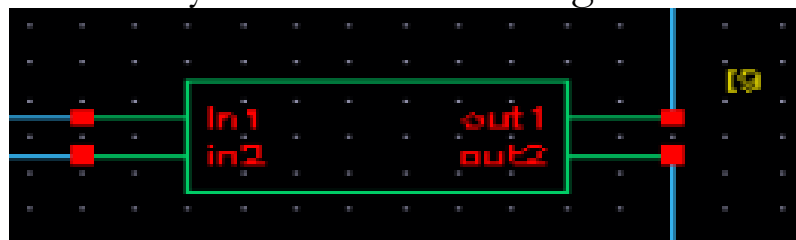
```
// VerilogA for thermal_sensor3, common_mode_shifter, veriloga

`include "constants.vams"
`include "disciplines.vams"

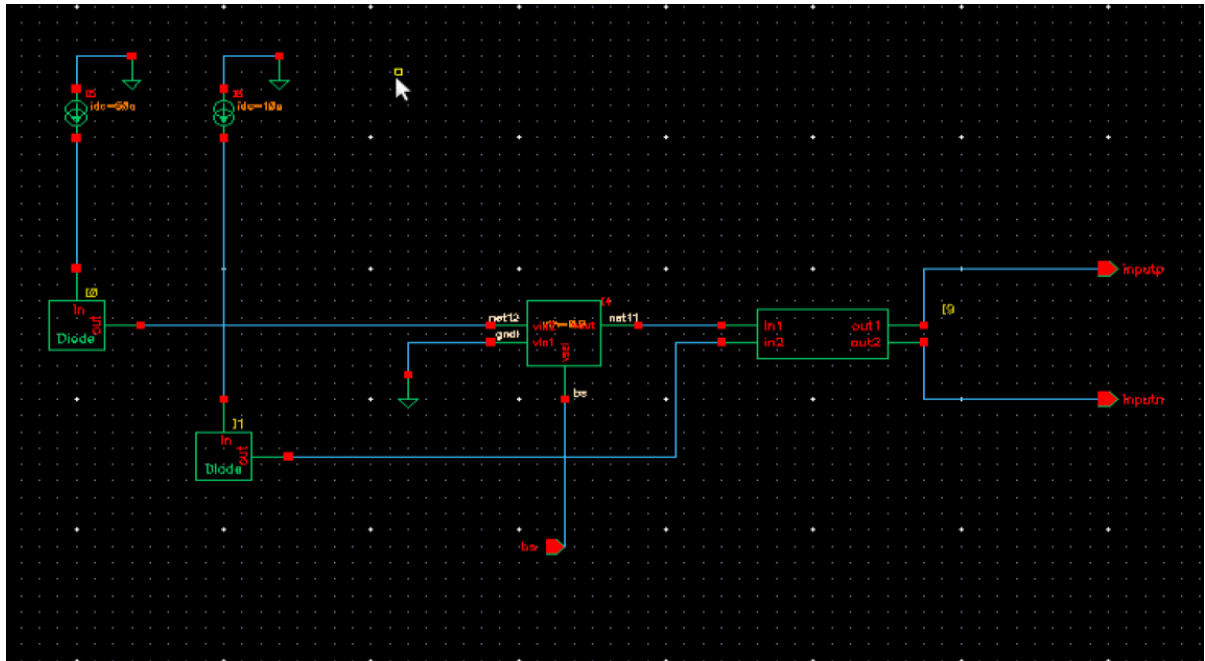
module common_mode_shifter(in1,in2,out1,out2);
input in1,in2;
output out1,out2;
electrical in1,in2,out1,out2;
parameter real common_mode = 900m from [0:inf];
real diff;

analog begin
    diff = V(in1) - V(in2);
    V(out1) <+ common_mode + (diff/2);
    V(out2) <+ common_mode - (diff/2);
end
endmodule
```

The symbol for this biasing circuit:



2.2.1.5 The Complete Schematic of the Sense Stage



The ratioed current densities produce different V_{BE} across the two thermal diode.

When the control wire of the analog MUX i.e. bs is equal to logic 0 then the double ended differential voltage between the two thermal diodes is selected which is ultimately the ΔV_{BE} that is required for charging of the integrating capacitor in the ADC.

If bs is equal to logic 1 then the analog selects the ground over the V_{BE} of one of the thermal diodes which ultimately gives out a double ended differential voltage of $-V_{BE}$ which is required for discharging the integrating capacitor in the ADC.

The biasing circuit biases the output of the analog MUX to maintain the input voltage to the ADC around the common-mode voltage (0.9V).

.2.2 The Delta-Sigma ADC

.2.2.1 Why Delta-Sigma Modulation?

Delta-Sigma Modulation reduces quantization noise by Oversampling and Noise Shaping techniques.

Quantization Noise:

The analog input signal can take any continuous value. But a digital n-bit signal can only settle to 2^n discrete values. It is this difference between the analog value and its digital representation, which causes the distortion known as the quantization noise. Quantization error is defined as a measure of an n-bit converter's failure to represent precisely an analog signal in the digital domain.

1. Oversampling, Noise shaping:

Oversampling is a process of sampling the input signal at a frequency much greater than the Nyquist frequency (Nyquist frequency, f_n is defined as twice the input signal bandwidth, f_b). This process greatly reduces the quantization noise in the required band. The sampling theorem or the basic Nyquist sample theory states that the sampling frequency of a signal must be at least twice the input signal frequency in order to reconstruct the sampled signal without distortion. Figure 2(a) shows the spectrum of an under sampled signal. Here the sampling frequency, f_s is less than twice the input signal frequency $2f_o$. The shaded portion of the figure shows the aliasing which occurs when the sampling theorem is not followed. We get a distorted signal at the output when a signal contaminated with aliasing is recovered. Figure 2(b) shows the spectrum of an oversampled signal. Here the sampled signal in the frequency domain appears as a series of band-limited signals at multiples of sampling frequency that are widely spaced. This process puts the entire input bandwidth at less than $f_s/2$ which reduces the aliasing.

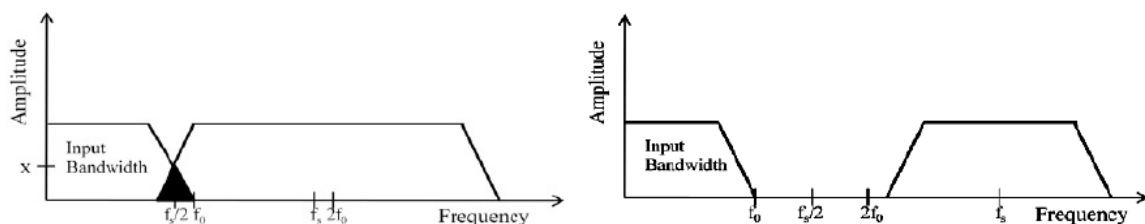


Figure 2(a): Under sampled signal 2(b) Oversampled signal spectrum

Noise shaping is a property of sigma-delta ADCs resulting from the application of feedback that extends dynamic range. A closed loop modulator works as a high-pass filter for quantization noise and as a low-pass filter for the input signal. When the signal is oversampled, the quantization noise power in the Nyquist bandwidth ($f_s/2$) spreads over the wider bandwidth, $Kf_s/2$ where; K is the oversampling ratio, which is shown in Figure 3. The total quantization noise is still the same but the quantization noise in the bandwidth of interest is greatly reduced. The figure also illustrates the noise shaping achieved by using the oversampled sigma-delta modulator. In a sigma-delta ADC, the analog modulator samples the input at oversampling ratio and after the input signal passes through the modulator it is fed into the digital filter or a decimator. The function of the digital filter is to provide a sharp cutoff at the bandwidth of interest, which essentially removes out of band quantization noise and signals as shown in Figure 3.

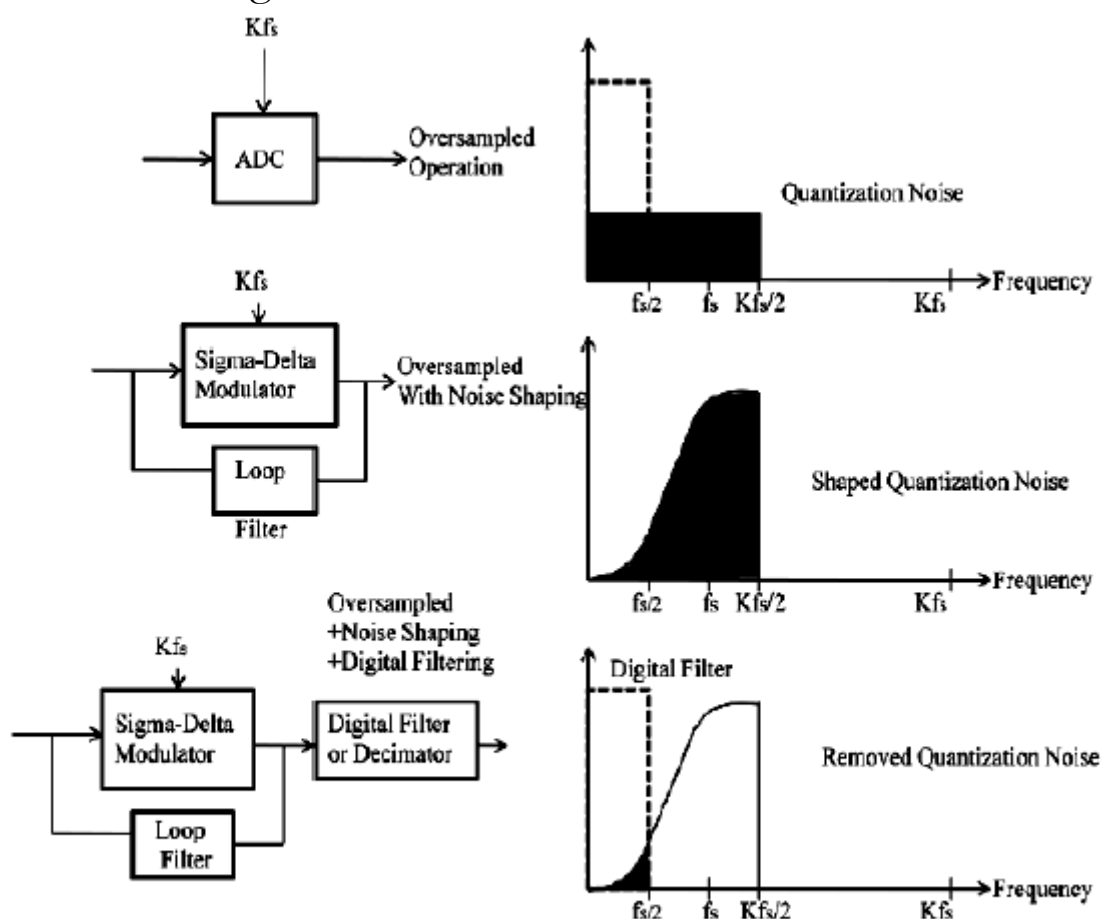
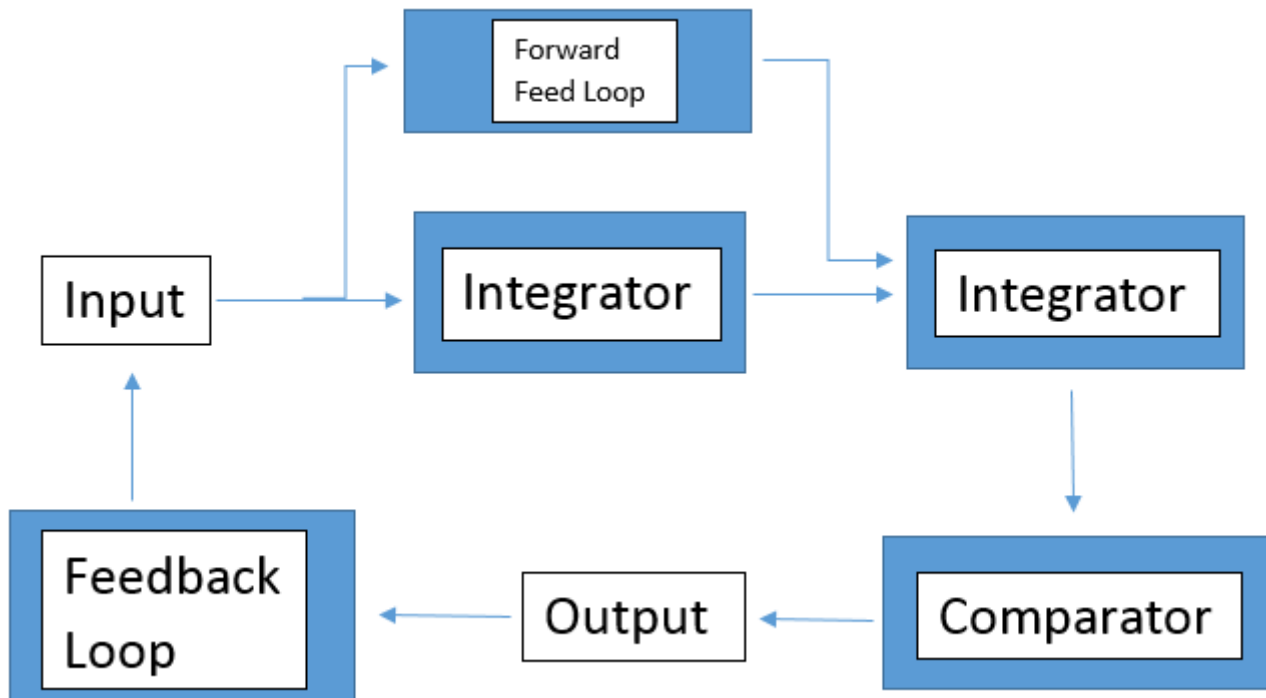


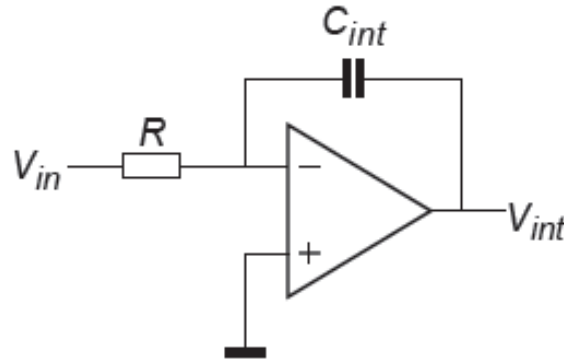
Figure 3: Effect of Noise shaping and Digital Filtering

.2.2.2 Stages in 2nd Order Delta-Sigma Analog to Digital Converter



The architecture of the delta-Sigma ADC is as shown above. The input ΔV_{BE} or $-V_{BE}$ is integrated by the first integrator. The integrator consists of an integrating capacitor which charges when the input is ΔV_{BE} and discharges if input is $-V_{BE}$. The output of the first integrator is then fed to the second integrator. The forward feed loop ensures that the response of the whole system with respect to a change in input is much faster. That ensures stability of the system. The output of the second integrator is fed to the comparator. When the positive edge of the clock happens and the input voltage of the comparator is more than zero then the comparator produces a pulse of logic 1. When the positive edge of the clock happens and the input voltage of the comparator is less than zero then the comparator produces a pulse of logic 0. This output is fed to the next stage of the thermal sensor i.e. the digital filter and as well as used as a feedback to select ΔV_{BE} when the output is logic zero and $-V_{BE}$ whenever the output is logic 1. This architecture basically encodes the analog input to a stream of ones and zeros. The density of ones in the bit stream is proportional to the input voltage to the ADC which is in turn proportional to temperature.

.2.2.3 Analog Integrator



The Input/output relation of the Analog Integrator is given by:

$$v_o = -\frac{1}{R_1 C_F} \int_0^t v_{in} dt$$

Where R_1 is the input resistance. C_F is the integrating Capacitor. Though this capacitor is very efficient performance-wise it takes up a lot of area because of the input resistor. A resistor takes up a lot of area than any other IC component. Therefore a switched capacitor network is used as a replacement to the resistor in the integrator circuit.

.2.2.4 Switched Capacitor Network

EQUIVALENT RESISTANCE OF A SWITCHED CAPACITOR CIRCUIT

Assume that $v_1(t)$ and $v_2(t)$ are changing slowly with respect to the clock period.

The average current is,

$$i_1(\text{average}) = \frac{1}{T} \int_0^T i_1(t) dt = \frac{1}{T} \int_0^{T/2} i_1(t) dt$$

Charge and current are related as,

$$i_1(t) = \frac{dq_1(t)}{dt}$$

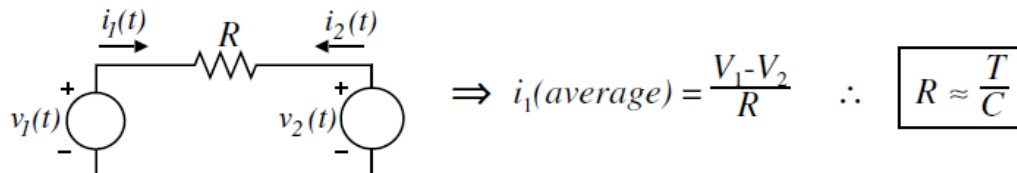
Substituting this in the above gives,

$$i_1(\text{average}) = \frac{1}{T} \int_0^{T/2} dq_1(t) = \frac{q_1(T/2) - q_1(0)}{T} = \frac{Cv_c(T/2) - Cv_c(0)}{T}$$

However, $v_c(T/2) = v_1(T/2)$ and $v_c(0) = v_2(0)$. Therefore,

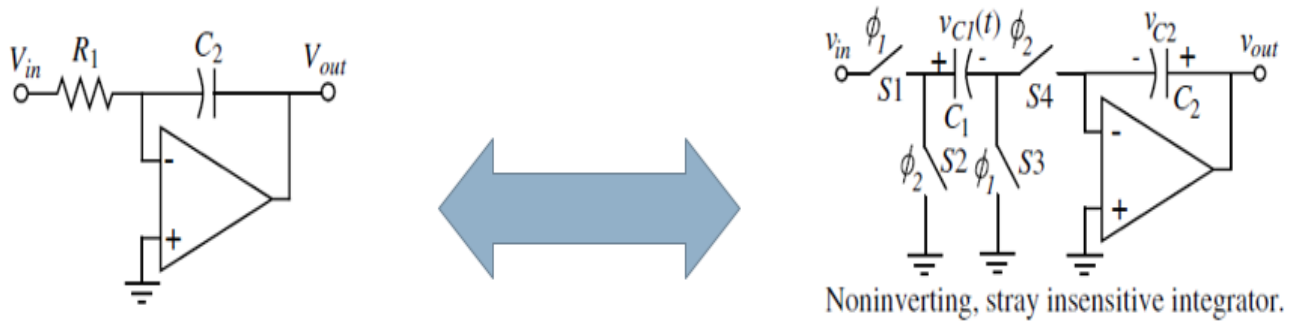
$$i_1(\text{average}) = \frac{C [v_1(T/2) - v_2(0)]}{T} \approx \frac{C [V_1 - V_2]}{T}$$

For the continuous time circuit:



For $v_1(t) \approx V_1$ and $v_2(t) \approx V_2$, the signal frequency must be much less than f_c .

.2.2.5 Switched Capacitor Integrator



Switched Capacitor Integrator with the above shown architecture is used in the Delta-sigma ADC.

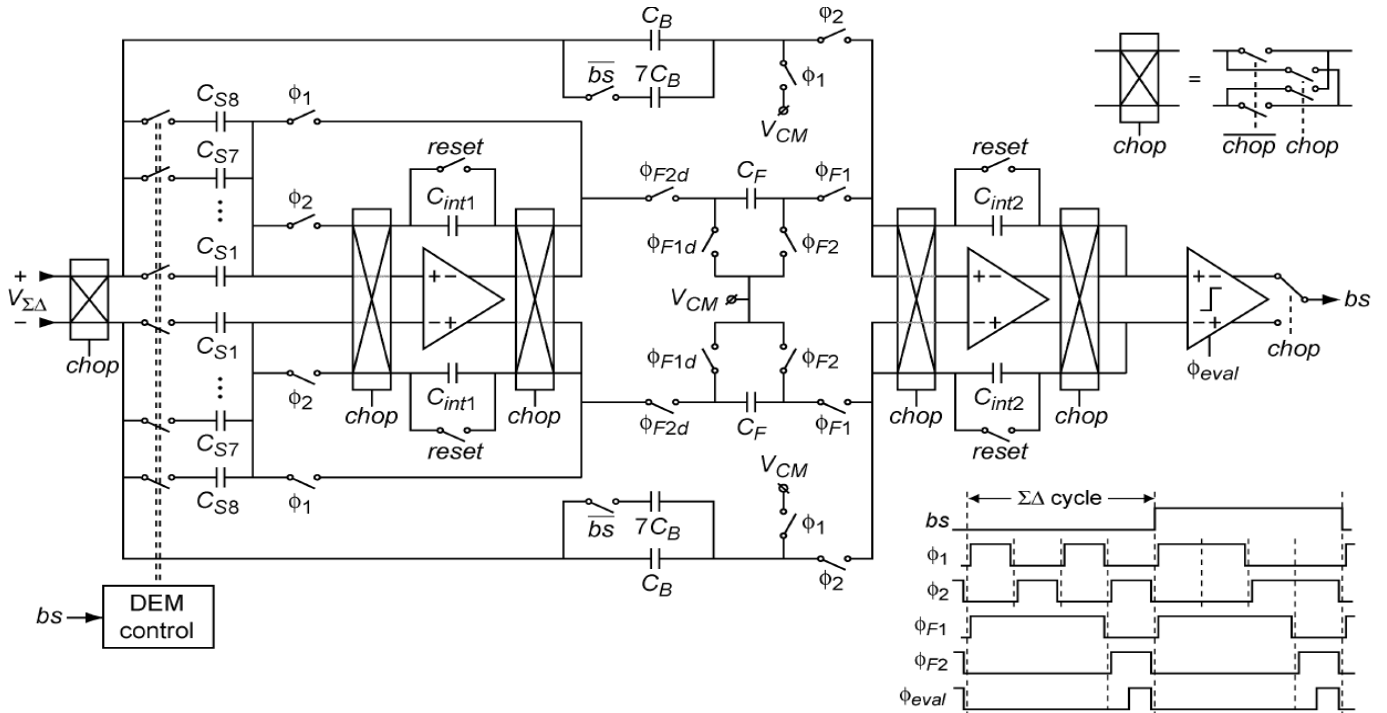
During the ϕ_1 phase switches S_1 and S_3 are closed. The other switches S_2 and S_4 are opened. As a result the sampling capacitor C_1 gets charged to V_{IN} .

During the ϕ_2 phase the switches S_2 and S_4 are closed and the switches S_1 and S_3 are opened. As a result the sampling capacitor gets discharged to zero potential. The charge released now appears on the integrating capacitor. Therefore the voltage that appears of the integrating capacitor is $\frac{C_1}{C_2} * V_{in}$.

Again in the next delta-sigma cycle the sampling capacitor charges and discharges and the integrating capacitor accumulates the charge discharged from the sampling capacitor.

If the Switch S_2 is connected to the negative terminal of the input V_{IN} instead of ground then the sampling capacitor discharges to $-V_{IN}$. In the ϕ_2 phase. This implies that the final voltage that appears on the integrating capacitor is $\frac{2C_1}{C_2} * V_{in}$

.2.2.6 The Delta-Sigma ADC circuit



This is a typical 2-stage differential Delta-Sigma Analog to Digital Converter.

A Chop is a network of switches which ensure that the negative and the positive electrical lines are used for opposite purpose half the time to ensure that the errors due to offset voltage of the amplifiers and the path dissimilarity be averaged out.

The clocks to the switched cap networks and the comparator is also shown above. $V_{\Sigma\Delta}$ is the input voltage to the ADC. bs is the bit stream output of the sigma-delta ADC. The value of $V_{\Sigma\Delta}$ is ΔV_{BE} or $-V_{BE}$ depending on whether bs is 0 or 1.

Since ΔV_{BE} is much smaller than $-V_{BE}$ in magnitude and the temperature coefficient of ΔV_{BE} is much lesser than that of $-V_{BE}$ we need to provide different integrator gains while integrating $-V_{BE}$ and ΔV_{BE} . As the temperature coefficient of ΔV_{BE} is about 16 times less than that of $-V_{BE}$. We try to compensate it by including another capacitor parallel to the sampling capacitor to boost the gain during ΔV_{BE} integration. We can also boost the gain by integrating the input voltage twice while V_{IN} is ΔV_{BE} .

.2.2.7 Verilog-A code for the Differential Amplifiers

```
// FUNCTION: Non-ideal OpAmp
// last revised: 22 Jan 2002 jbdavid
// REVISION HISTORY:
// Created BY: Jonathan David with modelwriter
// modified for Differential Operation
// VERSION: $Revision: 2.5 $
// AUTHORS: Modelwriter Standard Library
//          Cadence Design Systems, Inc.
//
// GENERATED BY: Affirma Modelwriter 2.23
//          ON: Tue Jan 22 20:55:47 PST 2002
//
// Description: Universal Opamp
// model opamp - Non Ideal OpAmp Model
// LIMITATIONS:
// INCLUDE FILES:

`include "discipline.h"
`include "constants.h"
//=====
=====

module DiffOpamp (vout_p, vout_n, vcmref, vin_p, vin_n, vspply_p,
vspply_n );
// PINS
    inout vin_p , vin_n;
//    vin_p - positive or non-inverting input
//    vin_n - negative or inverting input
    inout vspply_p, vspply_n; // pos and neg
    output vout_p, vout_n;
    input vcmref;
    electrical vin_p, vin_n, vout_p, vout_n, vcmref, vspply_p, vspply_n;
```

```

// INSTANCE PARAMETERS:
//      gain = Open loop voltage gain, or DC voltage gain
parameter real      gain = 1000.0 exclude 0.0;
//      ibias = Input bias current, the value is the same for
// both inputs [A]
parameter real      ibias = 0.0n;
//      pole_freq = Dominant pole frequency, or first corner
// frequency, eg. point where gain begins to roll of by 6 dB /
// octave [Hz]
parameter real pole_freq = 1.2;
//      rin = Differential input resistance, or resistance
// measured between both inputs [ohms]
parameter real      rin = 12.0M exclude 0.0;
//      rout = Single ended output resistance [ohms]
parameter real      rout = 75.0;
//      vin_offset = Input offset voltage, the voltage required for
// 0 volts output [V]
parameter real vin_offset = 0.0u;
//      vsoft = Output soft clipping point, measured from the
// supply rails [V]
parameter real      vsoft = 0.25;

```

```

// LOCAL VARIABLES: (Comment each one)

```

```

real c1, r1; // components of the dominant pole
real vin_val; // input diff voltage
real r_rout; // output resistance
real gm_nom; // first stage gm = gain/rout

```

```

// INTERNAL NODES

```

```

electrical coutp, coutn, vref;

```

```

// FUNCTIONS

```

```

//      {none}

```

```
//=====
=====
```

```
analog begin
```

```
  @(initial_step) begin // by default ALL analyses included (446+)
```

```
    r1 = gain;
```

```
    gm_nom = 1.0;
```

```
    c1 = 1/(`M_TWO_PI * pole_freq * gain);
```

```
    r_rout = rout;
```

```
  end
```

```
  vin_val= V(vin_p, vin_n) + vin_offset;
```

```
  // ----- Vref is at Virtual Ground
```

```
  V(vref, vsupply_n) <+ 0.5*V(vsupply_p,vsupply_n);
```

```
  // ----- Input Stage
```

```
  I(vin_p, vin_n) <+ vin_val / rin;
```

```
  I(vref, vin_p) <+ ibias;
```

```
  I(vref, vin_n) <+ ibias;
```

```
  // ----- GM stage
```

```
    I(vref, coutp) <+ gm_nom*vin_val ;
```

```
    I(vref, coutn) <+ -gm_nom*vin_val ;
```

```
  // ----- Dominant Pole.
```

```
  I(coutp, vref) <+ 2*c1*ddt(V(coutp, vref));
```

```
  I(coutp, vref) <+ 2*V(coutp, vref)/r1;
```

```
  I(coutn, vref) <+ 2*c1*ddt(V(coutn, vref));
```

```
  I(coutn, vref) <+ 2*V(coutn, vref)/r1;
```

```
  // ----- Output Stage.
```

```

I(vref, vout_p) <+ 2*V(coutp, vref)/r_rout;
I(vout_p, vref) <+ 2*V(vout_p, vref)/r_rout;
I(vref, vout_n) <+ 2*V(coutn, vref)/r_rout;
I(vout_n, vref) <+ 2*V(vout_n, vref)/r_rout;

```

```

// ----- Soft Output Limiting.
if (V(vout_p) > (V(vsupply_p) - vsoft))
    I(coutp, vref) <+ gm_nom*(V(vout_p, vsupply_p)+vsoft);
else if (V(vout_p) < (V(vsupply_n) + vsoft))
    I(coutp, vref) <+ gm_nom*(V(vout_p, vsupply_n)-vsoft);
// ----- Soft Output Limiting.
if (V(vout_n) > (V(vsupply_p) - vsoft))
    I(coutn, vref) <+ gm_nom*(V(vout_n, vsupply_p)+vsoft);
else if (V(vout_n) < (V(vsupply_n) + vsoft))
    I(coutn, vref) <+ gm_nom*(V(vout_n, vsupply_n)-vsoft);

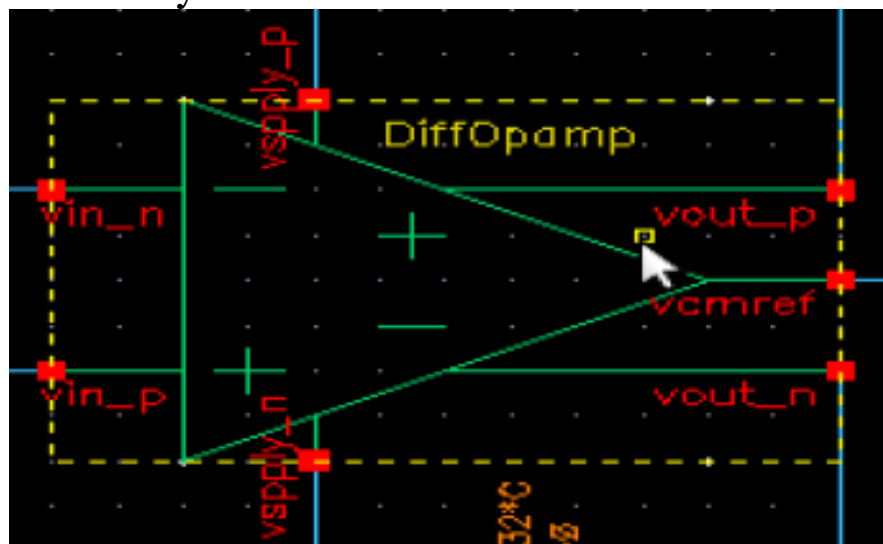
```

```

end
endmodule

```

The Symbol for this Behavioral Model:



.2.2.8 Verilog-A code for the Switches (Transmission Gates)

```
// FUNCTION: Switch with Risetime
// VERSION: $Revision: 2.6+ $
// AUTHORS: Modelwriter Standard Library
//          Cadence Design Systems, Inc.
//
// GENERATED BY: Affirma Modelwriter 2.23
//          ON: Mon May 05 10:58:16 EDT 2003
//
// Description: Non-Ideal Switch
// A switch with finite "on" and "off" resistances
// and nonzero transition time for switching between
// states. Impedance transition follows a logarithmic
// shift so that the impedance midway through the
// transition is the geometric mean of the on and off
// resistances.
// the switch changes state when the signal "control" is
// greater than the threshold voltage.
//
// PARAMETERS:
// roff = Switch Open Resistance [Ohms]
// ron = Switch Closed Resistance [Ohms]
// slack = time overstep considered negligible for threshold
// crossing sensing [S]
// tconv = Delay from threshold crossing to begining of
// resistance transition [S]
// tfall = Transition time from On to Off [S]
// trise = Transition time from Off to On [S]
// vth = Control signal threshold voltage [V]
// vtol = voltage overstep considered negligible for threshold
// crossing sensing [V]
//
`include "discipline.h"
`include "constants.h"
```

```

// sw_no Normally Open Switch
//

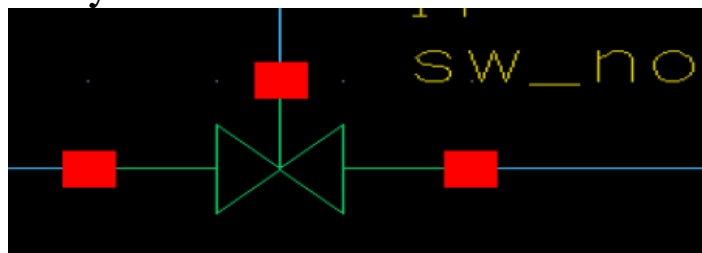
module sw_no (vin, control, vout);
  inout    vin, vout;
  input    control;
  voltage   control;
  electrical vin, vout;
  parameter real vth  = 1.5;
  parameter real ron  = 1.0 from (0:inf);
  parameter real roff = 10.0M from (ron:inf);
  parameter real tconv = 0.0n from [0:inf];
  parameter real trise = 4.0n from (0:inf);
  parameter real tfall = 4.0n from (0:inf);
  parameter real slack = 1n from (0:inf);
  parameter real vtol = 10.0m from (0:inf);
  real rout;
  integer rstate;

  analog begin

    rstate = (V(control) >= vth) ;
    // include point when crossing threshold in either direction:
    @(cross(V(control)-vth, 0, slack, vtol));
    // sweep resistance logarithmically through range:
    rout = roff*pow(ron/roff,transition(rstate, tconv, trise, tfall));
    I(vin,vout) <+ V(vin,vout)/rout;
  end
endmodule

```

The symbol for this behavioral model:



.2.2.9 The Verilog-A code for Comparator

```
//    FUNCTION: Clocked Comparator
// last revised: 23 Jan 2002 jbdavid
//
// GENERATED BY: Affirma Modelwriter 2.23
//            ON: Tue Jan 22 21:46:51 PST 2002
//
// Description: Clocked Comparator
//
//    An edge triggered comparator, with hysteresis,
//       and input offset. Configurable for active
//       high and active low enable.
//
// LIMITATIONS:
// INCLUDE FILES:
`include "discipline.h"
`include "constants.h"

// model CompLatched - Comparator
// enable signal is active: High
//
//=====
//=====

module CompLatched (vin_p, vin_n, enable, vsply_p, vsply_n, D,
D_);
// PINS
input vin_p, vin_n, enable;
output D, D_;
inout vsply_p, vsply_n; // pos and neg
electrical vsply_p, vsply_n; // pos and neg
electrical vin_p, vin_n, enable, D, D_;

// INSTANCE PARAMETERS:
//    hys = Input voltage threshold, after offset, required for
// output to change state [V]
```

```

//  n_off = Offset voltage added to vin_n pin [V]
//  one = Analog value for a Digital 1 on output [V]
//  p_off = Offset voltage added to vin_p pin [V]
//  slack = The smallest time interval considered negligible
// (abstol for clock) [S]
//  td = intrinsic delay from threshold cross to output
// change [S]
//  tfall = Transistion time for output fall [S]
//  trise = Transition time for output rise [S]
//  vth = Enable pin's threshold voltage [V]
//  zero = Analog value for a Digital 0 on output [V]
parameter real td = 50.0u from (0:inf);
parameter real hys = 25.0m from (0:inf);
parameter real p_off = 10.0u;
parameter real n_off = -10.0u;
parameter real trise = 20.0u from (0:inf);
parameter real tfall = 20.0u from (0:inf);
parameter real one = 3.0;
parameter real zero = 0.0;
parameter real vth = 1.5;
parameter real slack = 10.0p from (0:inf);
parameter integer init = 1;
parameter integer traceflag = 0;

// LOCAL VARIABLES: (Comment each one)
  real vin, halfhys, outstate;
// INTERNAL NODES
//  {none}
// FUNCTIONS
//  {none}

//=====
=====

analog begin
  @(initial_step) begin
    halfhys = hys/2.0;
    vin = V(vin_p) + p_off - V(vin_n) + n_off;

```

```

    outstate = (abs(vin) - abs(halfhys) > 0.0) ? zero : one ;

    if (traceflag) begin
        $strobe("%M at Init. D: %g D_: %g hys/2:%g", V(D), V(D_),
halfhys);
    end
end

vin = V(vin_p, vin_n) + p_off + n_off;

@(cross(vin - halfhys, +1, slack)) begin
    if (V(enable) > vth) begin
        outstate = one;
        if(traceflag)
            $strobe("%M at %g sec. output going high vin(p-n): %g
hys/2:%g",
                    $abstime, vin, halfhys);

    end
end

@(cross(vin + halfhys, -1, slack)) begin
    if (V(enable) > vth) begin
        outstate = zero;
        if(traceflag)
            $strobe("%M at %g sec. output going low vin(p-n): %g
hys/2:%g",
                    $abstime, vin, halfhys);

    end
end

// when re-enabled need to determine if condition has changed

@(cross(V(enable)-vth,+1)) begin // on rising edge
    if ((vin < -halfhys)&&(outstate > (one+zero)/2.0)) begin
        outstate = zero;
        if(traceflag)

```

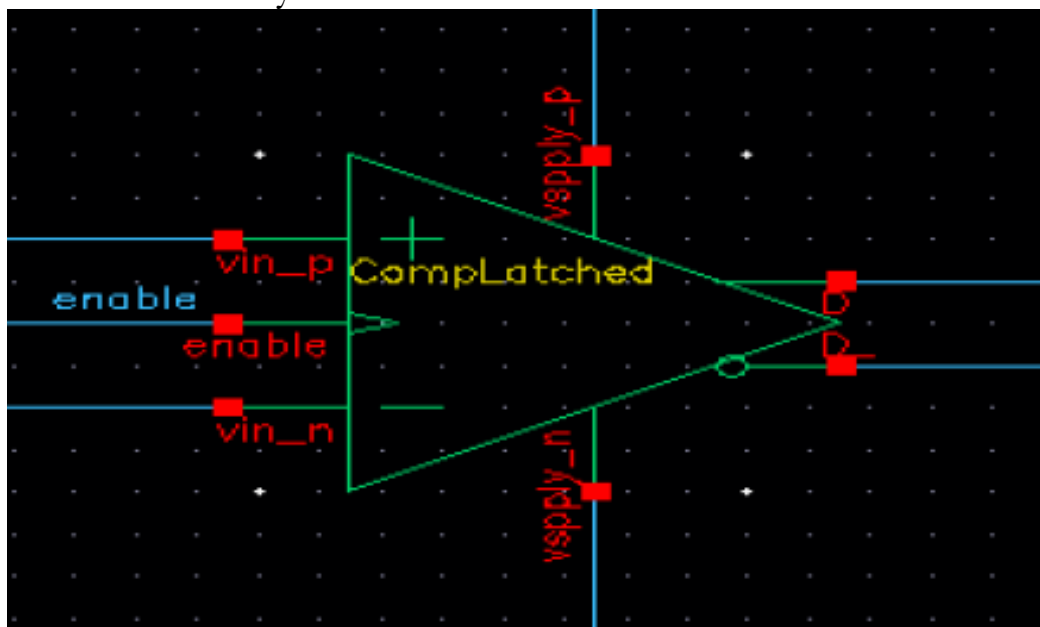
```

        $strobe("%M at %g sec. enable going high vin(p-n): %g
hys/2:%g",
                $abstime, vin, halfhys);
    end
    else if ((vin > halfhys)&&(outstate < (one+zero)/2.0)) begin
        outstate = one;
        if(traceflag)
            $strobe("%M at %g sec. enable going high vin(p-n): %g
hys/2:%g",
                    $abstime, vin, halfhys);
        end
    end
    // normal and complement outputs
    V(D) <+ transition (outstate, td, trise, tfall );
    V(D_) <+ one+zero-V(D);

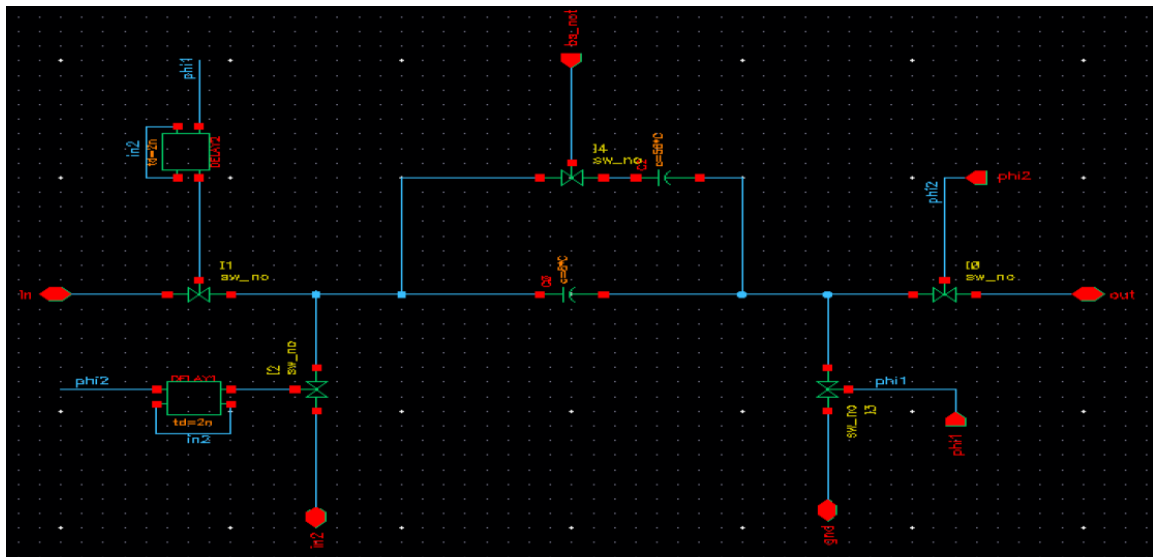
end
endmodule

```

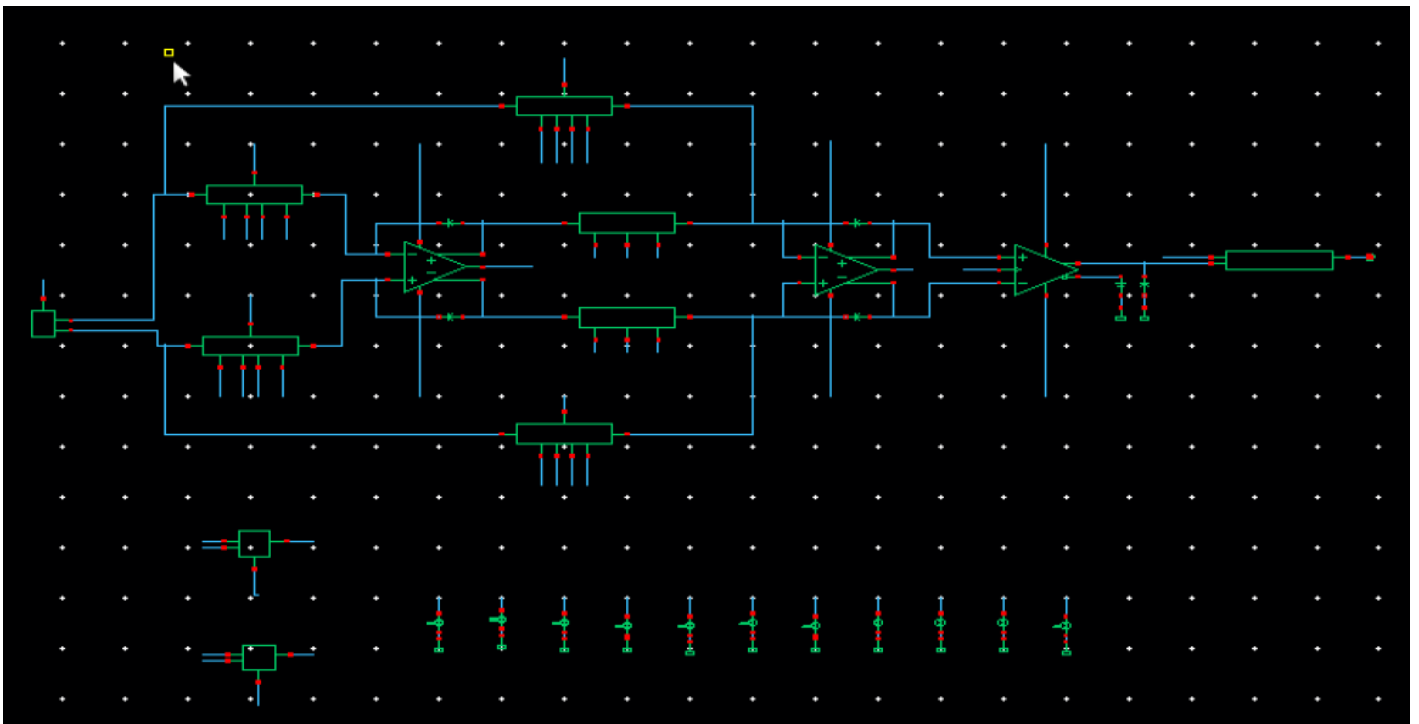
The symbol for this behavioral model:



.2.2.10 The schematic of the switched Capacitor Network



.2.2.11 The schematic of the complete Delta-Sigma Analog to Digital Converter



.2.3 Digital filter

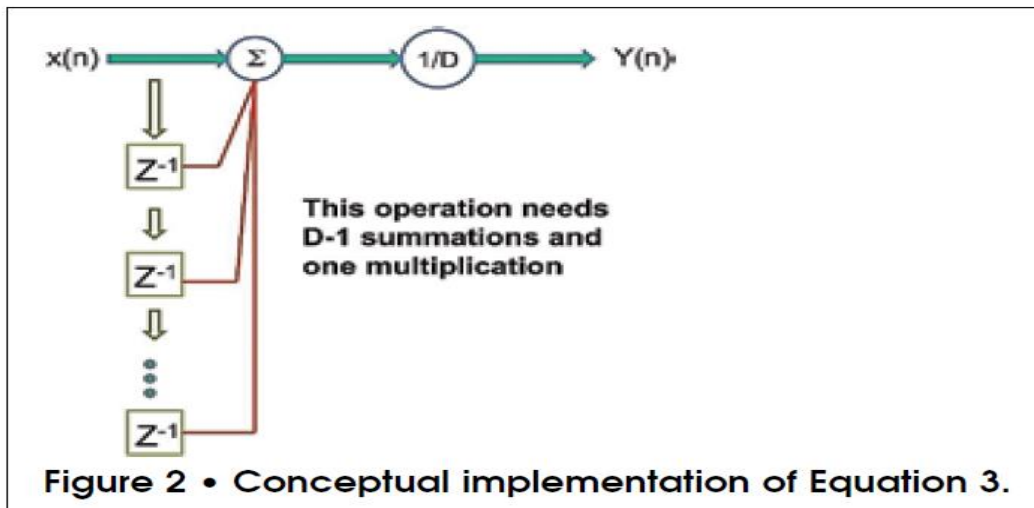
.2.3.1 Intro : IIR Filters vs FIR filters

1. The basic characteristics of Finite Impulse Response (FIR) filters are:
 - linear phase characteristic;
 - high filter order (more complex circuits); and
 - Stability.
2. The basic characteristics of Infinite Impulse Response (IIR) are:
 - non-linear phase characteristic;
 - low filter order (less complex circuits); and
 - Resulting digital filter has the potential to become unstable.

.2.3.2 The Transfer function of the Digital Filter

The basic functionality of the digital filter here is to

1. To decimate the bit stream from the ADC to decrease the bit rate so that it is process able in the further stages.
2. To calculate the moving average of the bit-stream which is basically proportional to the temperature.



Equation 4 can be simplified in the general form of Equations 5 and 6:

$$H(z) = [1 + z^{-1} + z^{-2} + z^{-3} + \dots + z^{1-D}] / D \quad \text{Eq. 5}$$

$$H(z) = \frac{1}{D} \sum_{0}^{D-1} z^{-n} \quad \text{Eq. 6}$$

Such an architecture requires $D \cdot \log_2 D$ number of memory units which takes a huge area and power. Therefore an alternate architecture which gives the same transfer function should be implemented.

.2.3.3 The Alternative Architecture

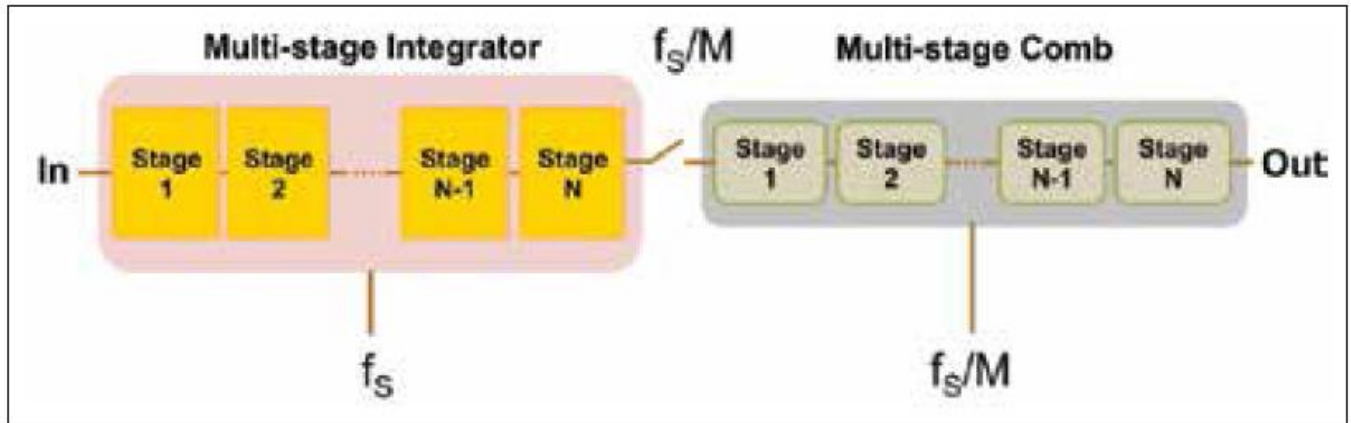


Figure 4 • Illustration of a multistage CIC filter.

The Transfer function of the above given architecture:

$$H(z) = H_{\text{int}}^K(z) H_{\text{comb}}^K(z) = \frac{(1 - z^{-DM})^K}{M^K (1 - z^{-1})^K}$$

This expression on expansion gives the transfer function we strive to achieve.

This architecture basically consists of a multiple integrators followed by same number of differentiators (comb) which act at a much lesser frequency. Some thumb rules while designing the digital filter.

- Number of stages of integrator should be equal to be number of stages of comb.
- The number of stages of Digital filter should be more than Number of Stages of the ADC.
- As the number of stages increases the sensitivity of the output/input increases.
- As the number of stages increase the non-linearity increases resulting in more complex Decimal Equivalent to Temperature Conversion Equation.

.2.3.4 Bit Width

We know that IIR filters take much lesser area than FIR filters. That is the exact reason for using this architecture. Digital Integrator is a IIR filter. Therefore for some inputs the output of the Integrator might be unstable. Therefore there will be data overflow. But to ensure that there is no data loss due to overflow we need to use the number of bit lines that is given by the below shown equation:

$$B_{out} = K * \log_2 (D * M) + B_{in}$$

Where K is the number of Stages in the Digital Filter.

D is the Decimation Factor.

M is the Feed-Forward Delay of the Comb filter.

B_{in} is the Bit width of the filter input.

B_{out} is the Bit width of the filter output.

.2.3.5 Implementation of Integrator and Differentiator

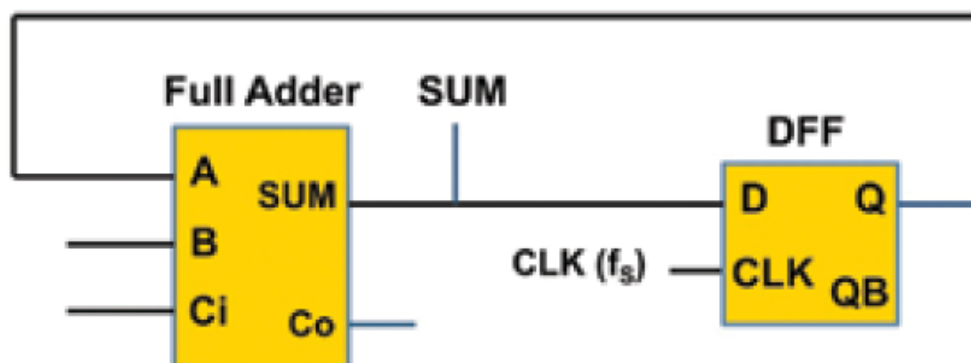


Figure 5 • Schematic of digital integrator.

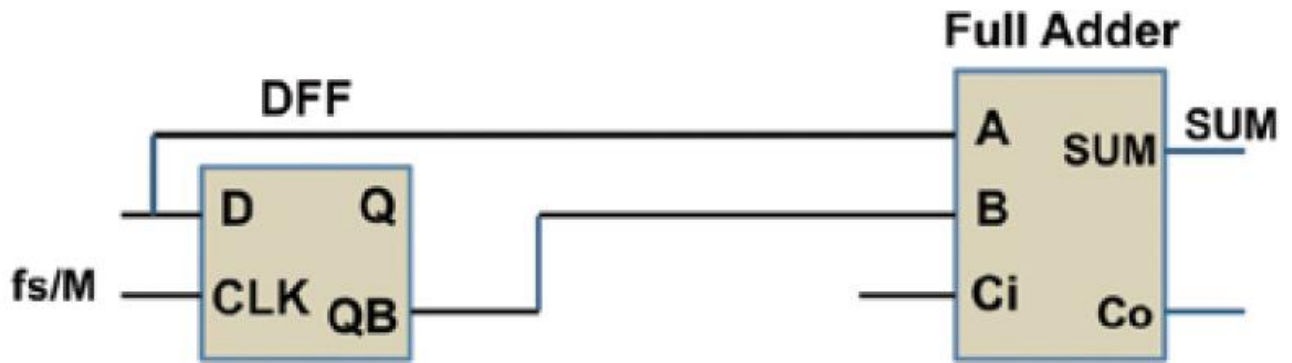


Figure 6 • Schematic of digital differentiator.

This is basically a 1-bit integrator and differentiator. But in practice we need Bout bit integrator and differentiator built from Bout bit registers and parallel or look-ahead adder/sub tractors. These architectures are just for better understanding only.

Anyway the effective number of Bits would not be equal to Bout. Some Bits have to be discarded for statistical and accuracy reasons. In this project the ENOB is 11 bits.

Anyway to build the behavioral model it is not necessary to understand the architecture of the digital filter. It can be implemented from the transfer function of the digital filter itself.

.2.3.6 Verilog-A code for the Digital filter

```
// VerilogA for intern, cicfilter, veriloga
```

```
`include "constants.vams"
`include "disciplines.vams"
```

```
// $ Date: 2015/11/19 11:05:30
//-----
// Decimator + CIC filter
// vin: [V,A]
// vout: [V,A]
// vclk: [V,A]
//
```

```
// INSTANCE parameters
// N          = Oversampling Ratio
// M          = Feed Forward Delay in the Comb Stage (Clock
Cycles)
// K          = Number of Stages in the CIC filter
// vtrans_clk = Transition voltage of the clock (V)
// tdel, trise, tfall = {usual} (s)
//
// MODEL parameters
// {none}
//
// Produces a cumulative average of 'N' samples of 'vin'.
// The average's power is raised by the Number of stages 'K' before
being outputted.
// 'vin' is sampled on positive 'vclk' transition.
// The cumulative average (raised to the power of 'K') of the previous set
of 'N' samples is outputted
// until a new set of 'N' samples has been captured.
// The output is  $K \cdot \log_2(N) + 1$  bits
//
// Transfer Function:  $(1/M \cdot (1 - Z^{-N}) / (1 - Z^{-1}))^K$ 
```

```
module cicfilter(vin,vout,vclk);
input vin,vclk;
output [0:24] vout;
electrical [0:24] vout;
electrical vin,vclk;
parameter integer N = 64 from [0:inf);
parameter integer K = 4 from [0:inf);
parameter integer M = 1 from [0:inf);
parameter real vtrans_clk = 0.9;
parameter real tdel = 0 from [0:inf);
parameter real trise = 0 from [0:inf);
parameter real tfall = 0 from [0:inf);
```

```
integer count,i;
integer sum;
real vout_val[0:24];
```

```

analog begin
    @(cross(V(vclk)-vtrans_clk,1.0)) begin
        sum = sum + V(vin)/1.8;
        count = count + 1;

        if (count == N) begin
            count = 0;
            sum = (sum/M)**K;
            while (i<25) begin
                vout_val[i] = 1.8*(sum % 2);
                sum = sum/2;
                i = i+1;
            end
        end
    end
end
generate j (0,24)begin
    V(vout[j]) <+ transition(vout_val[j],tdel,trise,tfall);
end
end
endmodule

```

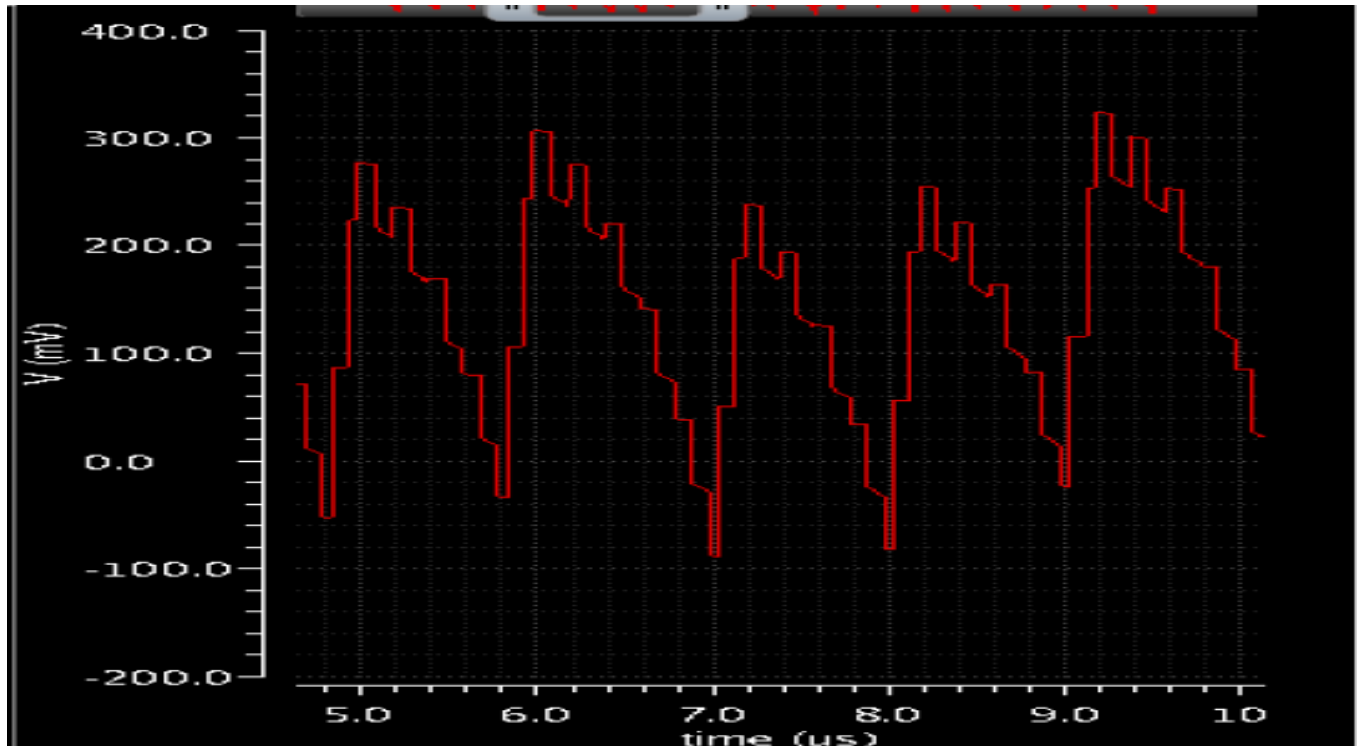
Symbol for this behavioral model:



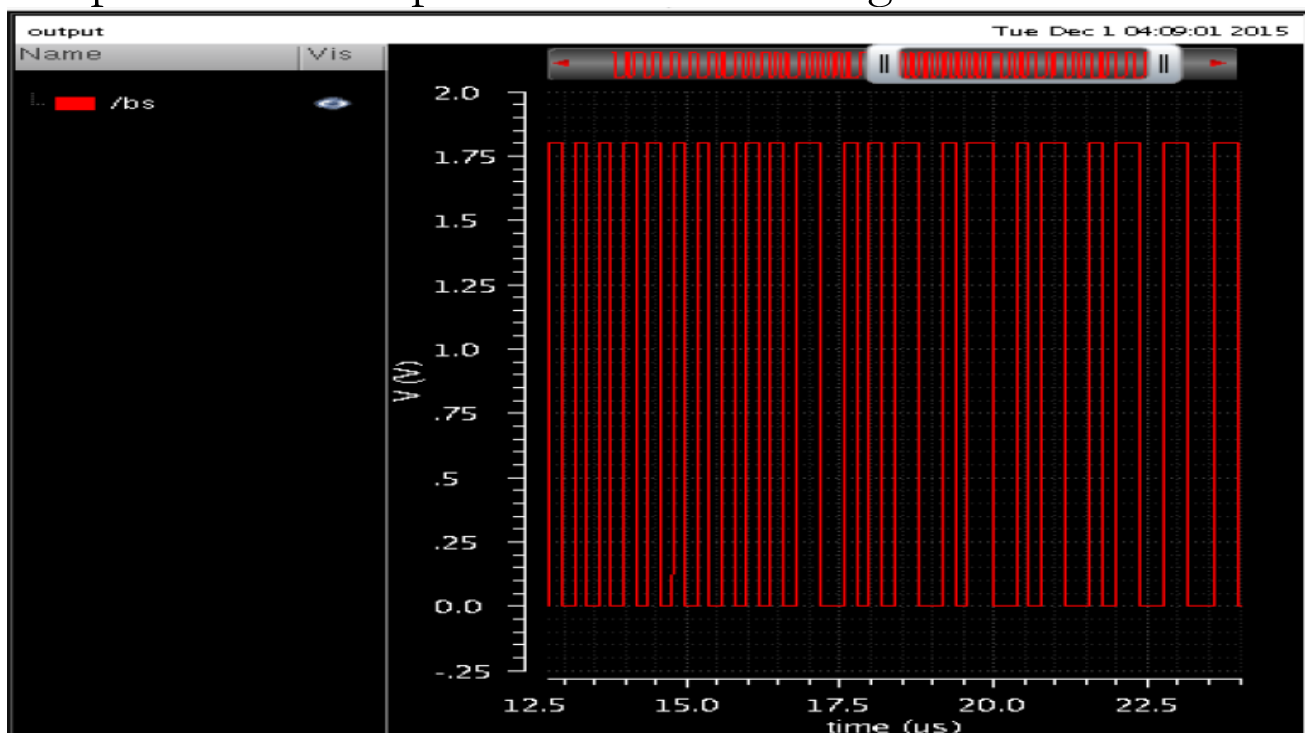
3.1 Results

At 350 K temperature:

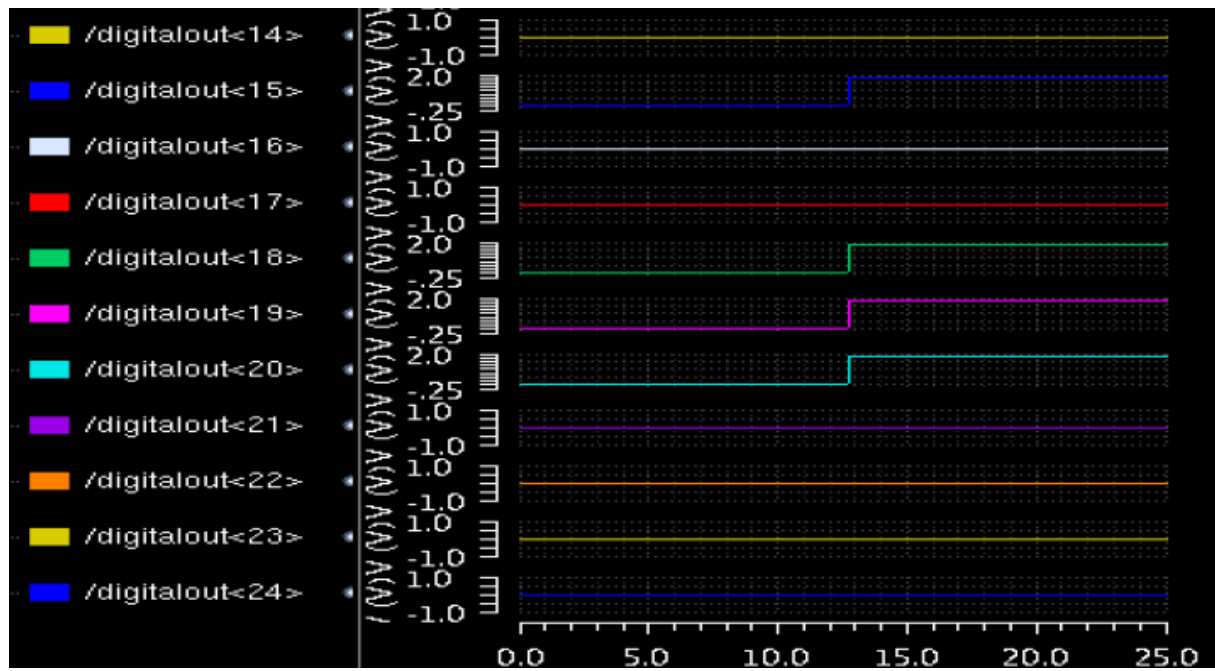
Output of the Second Integrator in the Delta-Sigma ADC.



Output of the Comparator in the Delta Sigma ADC.



Output from the digital filter (11 Bit stream)



3.2 The Decimal Equivalent Output of the ADC/Digital Filter

Temperature (Kelvin)	24	23	22	21	20	19	18	17	16	15	14	Decimal Number
100	0	0	0	0	0	0	0	0	0	0	0	0
110	0	0	0	0	0	0	0	0	0	0	1	1
120	0	0	0	0	0	0	0	0	0	0	1	1
130	0	0	0	0	0	0	0	0	0	1	1	3
140	0	0	0	0	0	0	0	0	0	1	1	3
150	0	0	0	0	0	0	0	0	1	0	1	5
160	0	0	0	0	0	0	0	0	1	0	1	5
170	0	0	0	0	0	0	0	0	1	1	1	7
180	0	0	0	0	0	0	0	1	0	0	1	9
190	0	0	0	0	0	0	0	1	0	1	1	11
200	0	0	0	0	0	0	0	1	1	1	0	14
210	0	0	0	0	0	0	1	0	0	0	1	17
220	0	0	0	0	0	0	1	0	1	0	0	20
230	0	0	0	0	0	0	1	0	1	1	1	23
240	0	0	0	0	0	0	1	1	0	1	1	27
250	0	0	0	0	0	1	0	0	0	0	0	32
260	0	0	0	0	0	1	0	0	1	0	1	37
270	0	0	0	0	0	1	0	1	0	1	1	43
280	0	0	0	0	0	1	1	0	0	0	1	49
290	0	0	0	0	0	1	1	1	0	0	0	56
300	0	0	0	0	1	0	0	0	0	0	0	64
310	0	0	0	0	1	0	0	1	0	0	0	72
320	0	0	0	0	1	0	1	0	0	0	1	81
330	0	0	0	0	1	0	1	1	0	1	1	91
340	0	0	0	0	1	1	0	0	1	1	0	102
350	0	0	0	0	1	1	1	0	0	1	0	114
360	0	0	0	0	1	1	1	1	1	1	1	127
370	0	0	0	1	0	0	0	1	1	0	1	141
380	0	0	0	1	0	0	1	1	1	0	0	156
390	0	0	0	1	0	1	0	1	1	0	0	172
400	0	0	0	1	0	1	1	1	1	0	1	189
410	0	0	0	1	1	0	1	0	0	0	0	208
420	0	0	0	1	1	1	0	0	1	0	0	228
430	0	0	0	1	1	1	1	1	0	1	0	250
440	0	0	1	0	0	0	1	0	0	0	1	273
450	0	0	1	0	0	1	0	1	0	0	1	297
460	0	0	1	0	1	0	0	0	1	0	0	324
470	0	0	1	0	1	0	1	1	1	1	1	351
480	0	0	1	0	1	1	1	1	1	0	1	381
490	0	0	1	1	0	0	1	1	1	0	0	412

By applying polynomial fit on the data we can easily get the relation between the temperature and the decimal equivalent of the Output from the ADC/digital Filter.

3.3 Conclusion

The Decimal Equivalent of 11-Bit output from the 4-stage CIC filter of the 2-stage Sigma Delta ADC to Temperature Conversion Equation:

- $T = 265 + 5\sqrt{(49 + 8(D - 64))}$.
- Non-Linearity due to design of the 4-stage CIC filter instead of 3-stage CIC filter.

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